

Jeongsang Ryu

PhD Candidate, Mechanical Engineering

UNIST, High-Assurance Mobility and Control Lab

Ulsan, South Korea

✉ ryujs@unist.ac.kr

🌐 jeongsang-ryu.github.io

🔗 jeongsang-ryu

jeongsangryu@gmail.com

Research Interests

SLAM, LiDAR-Inertial Odometry, Autonomous Racing, End-to-End Autonomous Driving, Reinforcement Learning, Robot Perception

Education

Mar 2024 – **Combined MS/PhD, Mechanical Engineering, UNIST**

Feb 2030 (expected) High-Assurance Mobility and Control Lab (HMCL) — Research: SLAM, LiDAR-Inertial Odometry, RL for Autonomous Racing

Mar 2018 – **BS, Mechanical Engineering, UNIST**

Feb 2024 GPA: 3.75 / 4.3

Teaching

2026 **Instructor, 2026 IFAC RoboRacer Bootcamp**

Gallery, Tutorial materials

Mar 2026 – **Teaching Assistant — MEN472: Creating Autonomous Car, UNIST, Mechanical**

Jun 2026 *Engineering*

Mar 2025 – **Teaching Assistant — MEN472: Creating Autonomous Car, UNIST, Mechanical**

Jun 2025 *Engineering*

Mar 2024 – **Teaching Assistant — MEN491: Creating Autonomous Car, UNIST, Mechanical**

Jun 2024 *Engineering*

2023 **Teaching Assistant, 2023 Korea F1TENTH Tutorial**

Gallery

Service

2026 **Referee, Organizer, 29th RoboRacer Autonomous Racing Competition**

IFAC, Busan

2026 **Referee, Organizer, RoboRacer PNU-Hyunsong Championship**

Busan

2025 **Referee, Organizer, 4th F1TENTH Korea Championship**

ICCAS, Incheon

2024 **Referee, Organizer, 3rd F1TENTH Korea Championship**

ICCAS, Jeju

2024 **Referee, 18th RoboRacer Autonomous Racing Competition**

IV, Jeju

Experience

Aug 2023 – **Undergraduate Research Intern, High-Assurance Mobility and Control Lab, UNIST**

Feb 2024 Autonomous racing algorithm development on F1TENTH platform. Motion planning, control, and ROS-based simulation and real vehicle testing.

Mar 2021 – **Undergraduate Research Intern, Clean Combustion & Energy Research Lab, UNIST**

Nov 2022 Neural network models for chemical kinetics simulation. Flame stabilization and extinction research using ML (Python, TensorFlow, OpenFOAM).

Apr 2019 – **Military Service**, *Republic of Korea Army*
Nov 2020

Skills

Languages C++, Python, Swift
Frameworks ROS / ROS2, Isaac Sim, Isaac Lab, ARKit, PyTorch, Docker
& Tools
Platforms Linux, iOS

Awards

2025 2nd Place, 24th RoboRacer Autonomous Racing Competition (ICRA, Atlanta, Team Leader)
2024 8th Place, 2024 MOTIE Competition (GNSS + IMU state estimation, Integration with LiDAR-based state estimation)
2023 1st Place, 2nd F1TENTH Korea Championship (ICCAS, Yeosu, Localization, Raceline Optimization)

Publications

To be updated.